

Monitoring Task -Sudir

Task Description:

Install Prometheus and Grafana on a Linux EC2 machine, connect Prometheus to Grafana, and create a dashboard to view metrics.

Techstacks needs to be used :

- **AWS EC2**
- **Prometheus**
- **Grafana**
- **Node Exporter**

How do I submit my work?

- **Push all your work files to GitHub (O/P screenshot images must).**
- **Submit your URLs in the portal.**

Terms and Conditions?

- **You agree to not share this confidential document with anyone.**
- **You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.**
- **We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.**

NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.

Step 1 - Install Node Exporter

```
cd /opt
```

```
sudo useradd --no-create-home --shell /bin/false node_exporter
```

```
wget
```

```
https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-1.9.1.linux-amd64.tar.gz
```

```
tar -xvf node_exporter-1.9.1.linux-amd64.tar.gz
```

```
sudo cp node_exporter-1.9.1.linux-amd64/node_exporter /usr/local/bin/
```

```
root@ip-172-31-41-142 terraform-ec2]# cd /opt/
root@ip-172-31-41-142 opt]# sudo useradd --no-create-home --shell /bin/false node_exporter
root@ip-172-31-41-142 opt]# wget https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_
-2026-06-04 10:10:37-- https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-
esolving github.com (github.com)... 20.207.73.82
connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response... 302 Found
location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/dc8ec09c-2975
ttachment%3B+filename%3Dnode_exporter-1.9.1.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d4
3A58%3A052&sk=2026-06-04T10%3A58%3A15Z&sk=bc&skv=2018-11-09&sig=nQHfHRbFc4POiG1at%2Bbis3Kh0AEkkiNODWG%2BWF
2WF%2S1hc3NldHMuZ210aHVidXNlcmNvbniRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4MDU2OTYaNywiYmJmIjoxNzgWNTY3ODM
xik5CFvSLOhlwRwQlkMwrQBTqIc&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.9.1.lin
-2026-06-04 10:10:37-- https://release-assets.githubusercontent.com/github-production-release-asset/952405
58%3A15Z&rsct=attachment%3B+filename%3Dnode_exporter-1.9.1.linux-amd64.tar.gz&rsct=application%2Foctet-stre
=2026-06-04T09%3A58%3A05Z&sk=2026-06-04T10%3A58%3A15Z&sk=bc&skv=2018-11-09&sig=nQHfHRbFc4POiG1at%2Bbis3Kh0
iwiYXVtIjoicmV5ZWZ%2S1hc3NldHMuZ210aHVidXNlcmNvbniRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4MDU2OTYaNywiYmJmIj
5xRAYfBz1Cf9zRFxik5CFvSLOhlwRwQlkMwrQBTqIc&response-content-disposition=attachment%3B%20filename%3Dnode_exp
esolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133, 18
connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.111.133|:4
HTTP request sent, awaiting response... 200 OK
length: 11582410 (11M) [application/octet-stream]
saving to: 'node_exporter-1.9.1.linux-amd64.tar.gz'

node_exporter-1.9.1.linux-amd64.tar.gz          100%[=====]

2026-06-04 10:10:38 (108 MB/s) - 'node_exporter-1.9.1.linux-amd64.tar.gz' saved [11582410/11582410]

root@ip-172-31-41-142 opt]# tar -xvf node_exporter-1.9.1.linux-amd64.tar.gz
node_exporter-1.9.1.linux-amd64/
node_exporter-1.9.1.linux-amd64/LICENSE
node_exporter-1.9.1.linux-amd64/NOTICE
node_exporter-1.9.1.linux-amd64/node_exporter
root@ip-172-31-41-142 opt]# sudo cp node_exporter-1.9.1.linux-amd64/node_exporter /usr/local/bin/
```

Create service:

```
sudo vi /etc/systemd/system/node_exporter.service
```

[Unit]

Description=Node Exporter

After=network.target

[Service]

User=node_exporter

Group=node_exporter

Type=simple

ExecStart=/usr/local/bin/node_exporter

[Install]

WantedBy=multi-user.target

Start:

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable node_exporter
```

```
sudo systemctl start node_exporter
```

Verify:

```
sudo systemctl status node_exporter
```

Test:

```
curl localhost:9100/metrics
```

```
[root@ip-172-31-41-142 opt]# sudo vim /etc/systemd/system/node_exporter.service
[root@ip-172-31-41-142 opt]# sudo systemctl daemon-reload
[root@ip-172-31-41-142 opt]# sudo systemctl enable node_exporter
Created symlink /etc/systemd/system/multi-user.target.wants/node_exporter.service -> /etc/systemd/system/node_exporter.service.
[root@ip-172-31-41-142 opt]# sudo systemctl start node_exporter
[root@ip-172-31-41-142 opt]# curl localhost:9100/metrics
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0
go_gc_duration_seconds_sum 0
go_gc_duration_seconds_count 0
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise 100. This value is set by the GOGC environment variable.
# TYPE go_gc_gogc_percent gauge
```

Step 2 - Install Prometheus

cd /opt

```
sudo useradd --no-create-home --shell /bin/false prometheus
```

```
wget https://github.com/prometheus/prometheus/releases/download/v3.5.0/prometheus-3.5.0.linux-amd64.tar.gz
```

```
tar -xvf prometheus-3.5.0.linux-amd64.tar.gz
```

```
sudo mkdir -p /etc/prometheus
```

```
sudo cp prometheus-3.5.0.linux-amd64/prometheus /usr/local/bin/
```

```
sudo cp prometheus-3.5.0.linux-amd64/promtool /usr/local/bin/
```

```
sudo cp -r prometheus-3.5.0.linux-amd64/consoles /etc/prometheus/
```

```
sudo cp -r prometheus-3.5.0.linux-amd64/console_libraries /etc/prometheus/
```

Create configuration:

```
sudo vi /etc/prometheus/prometheus.yml
```

global:

- scrape_interval: 15s

scrape_configs:

- job_name: "prometheus"

- static_configs:

- targets: ["localhost:9090"]

- job_name: "node_exporter"

- static_configs:

- targets: ["localhost:9100"]

Create service:

```
sudo vi /etc/systemd/system/prometheus.service
```

[Unit]

Description=Prometheus

After=network.target

[Service]

User=prometheus

Group=prometheus

ExecStart=/usr/local/bin/prometheus \

--config.file=/etc/prometheus/prometheus.yml

Restart=always

[Install]

WantedBy=multi-user.target

Start:

sudo systemctl daemon-reload

sudo systemctl enable prometheus

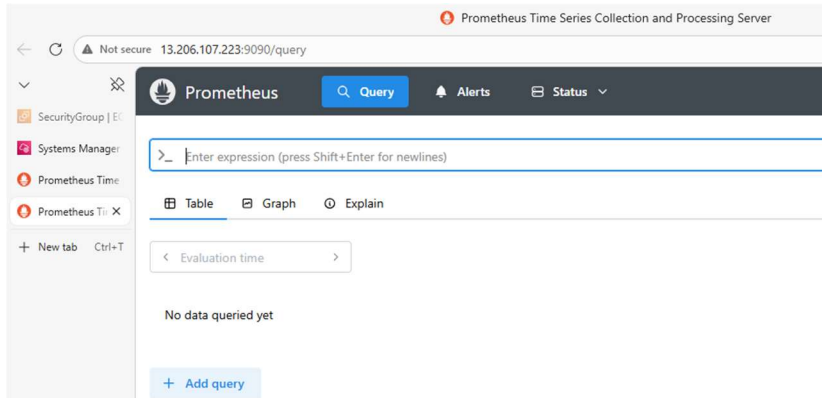
sudo systemctl start prometheus

Verify:

sudo systemctl status Prometheus

Browser:

<http://13.206.107.223:9090>



Step 3 - Install Grafana

```
sudo tee /etc/yum.repos.d/grafana.repo <<EOF
[grafana]
name=grafana
baseurl=https://rpm.grafana.com
repo_gpgcheck=1
enabled=1
gpgcheck=1
gpgkey=https://rpm.grafana.com/gpg.key
sslverify=1
EOF
```

Install:

```
sudo dnf install grafana -y
```

Start:

```
sudo systemctl enable grafana-server
sudo systemctl start grafana-server
```

```

[root@ip-172-31-41-142 opt]# sudo dnf install grafana -y
grafana
grafana
Importing GPG key 0x10458545:
  Userid : "Grafana Labs <engineering@grafana.com>"
  Fingerprint: 853A E77B ADB6 30A6 8304 6005 963F A277 1045 8545
  From : https://rpm.grafana.com/gpg.key
grafana
Last metadata expiration check: 0:00:18 ago on Thu Jun 4 10:32:33 2026.
Dependencies resolved.
=====
Package                               Architecture
-----
Installing:
grafana                               x86_64
=====
Transaction Summary
-----
Install 1 Package

Total download size: 212 M
Installed size: 991 M
Downloading Packages:
grafana_13.0.2_26816849631_linux_amd64.rpm
=====

```

Verify:

`sudo systemctl status grafana-server`

```

root@ip-172-31-41-142 opt]# sudo systemctl enable grafana-server
udo systemctl start grafana-server
reated symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service -> /usr/lib/sy
root@ip-172-31-41-142 opt]# sudo systemctl status grafana-server
grafana-server.service - Grafana instance
  Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: disabled)
  Active: active (running) since Thu 2026-06-04 10:34:34 UTC; 7s ago
    Docs: http://docs.grafana.org
   Main PID: 30466 (grafana)
      Tasks: 19 (limit: 1120)
     Memory: 463.5M
        CPU: 4.031s
    CGroup: /system.slice/grafana-server.service
            └─30466 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --p
              └─30471 /usr/share/grafana/data/plugins-bundled/elasticsearch/gpx_grafana_elasticse

Jun 04 10:34:38 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=plugin.backg
Jun 04 10:34:39 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=plugin.insta
Jun 04 10:34:39 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=installer.fs
Jun 04 10:34:39 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=plugins.regi
Jun 04 10:34:39 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=plugin.backg
Jun 04 10:34:39 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=plugin.insta
Jun 04 10:34:40 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=installer.fs
Jun 04 10:34:41 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=plugins.regi
Jun 04 10:34:41 ip-172-31-41-142.ap-south-1.compute.internal grafana[30466]: logger=plugin.backg
lines 1-22/22 (END)
root@ip-172-31-41-142 opt]# ss -tnlp | grep 3000
LISTEN 0      4096          *:3000          *:*    users: (("grafana",pid=30466,fd=18))
root@ip-172-31-41-142 opt]#

```

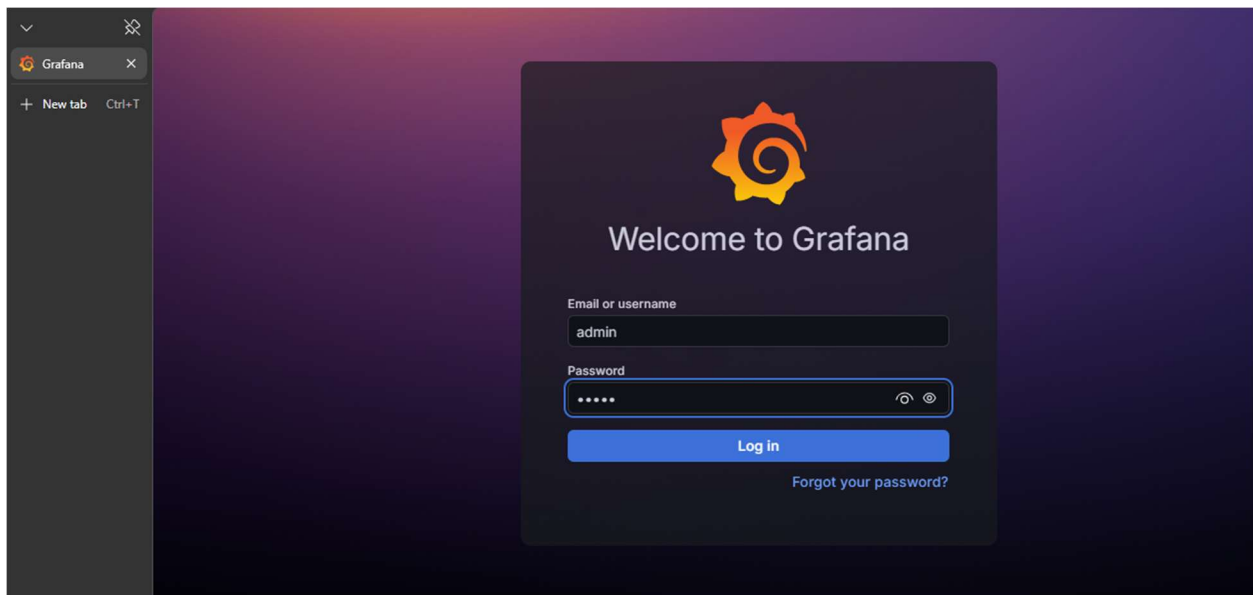
Browser:

<http://13.206.107.223:3000>

Default login:

admin

admin



Step 4 - Add Prometheus Data Source

Grafana:

Connections

→ Data Sources

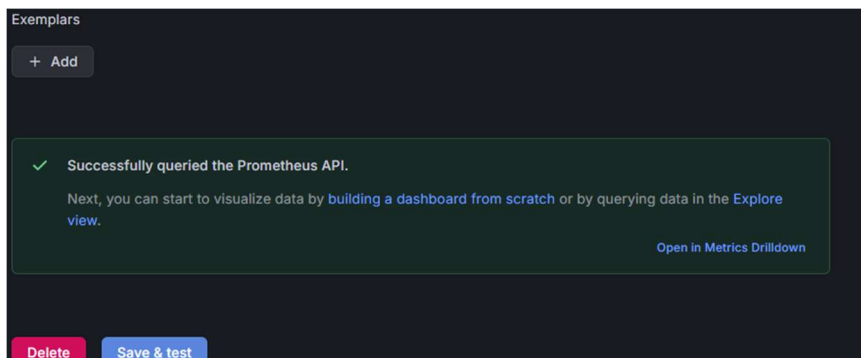
→ Add Data Source

→ Prometheus

URL:

`http://localhost:9090`

Save & Test.



Step 5 - Create Dashboard

Grafana:

Dashboards

→ New

→ Import

Use dashboard ID:

1860

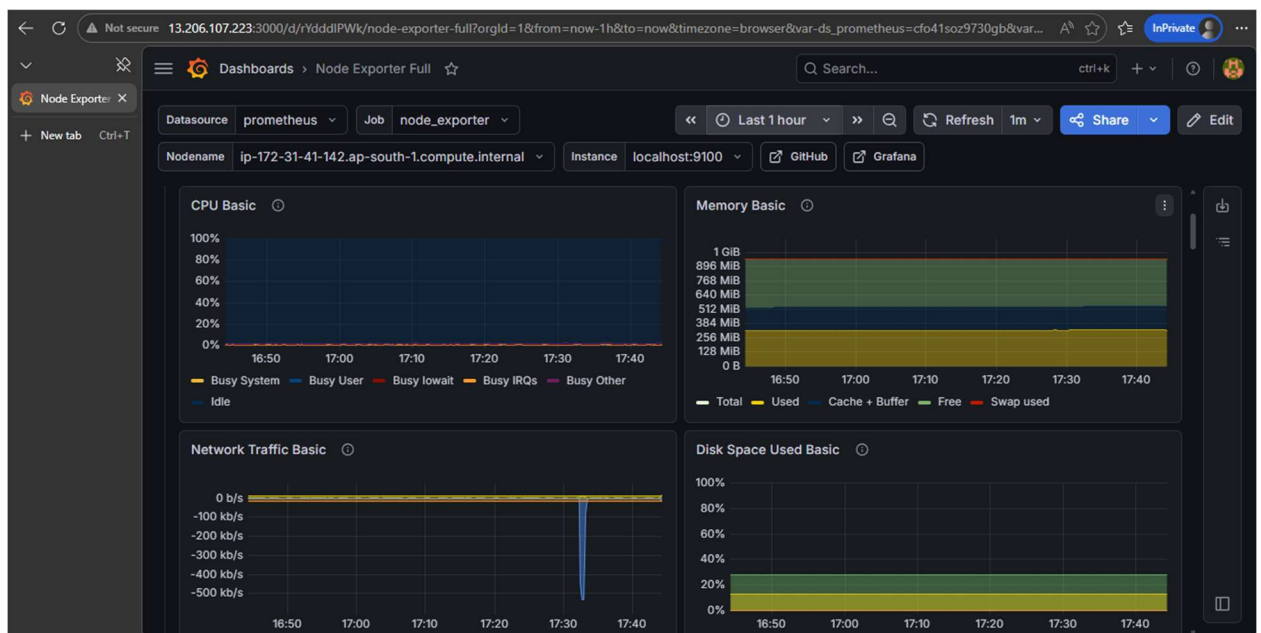
This is the popular Node Exporter Full dashboard.

Select:

Prometheus

Import.

- CPU Usage
- Memory Usage
- Disk Usage
- Network Usage



Security Group

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interface

sg-095d9c08138ec8bff - launch-wizard-1

Inbound security group rules successfully modified on security group sg-095d9c08138ec8bff (launch-wizard-1)

Details

sg-095d9c08138ec8bff - launch-wizard-1

Actions

Details

Security group name

launch-wizard-1

Security group ID

sg-095d9c08138ec8bff

Description

launch-wizard-1 created 2026-06-04T09:02:49.12Z

VPC ID

vpc-08cd654ebd17167be

Owner

645437362580

Inbound rules count

4 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (4)

Search

Manage tags

Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-08ac1914dbe75c659	IPv4	HTTPS	TCP	443	0.0.0.0/0	-
☐	-	sgr-0021d365f487b823e	IPv4	Custom TCP	TCP	9090	0.0.0.0/0	prometheus
☐	-	sgr-0e32899a363390343	IPv4	Custom TCP	TCP	3000	0.0.0.0/0	grafana
☐	-	sgr-0658dd6aab27a11e1	IPv4	SSH	TCP	22	0.0.0.0/0	-